

SOFTWARE TEST PLAN & VERIFICATION SPECIFICATION

Test Plan, Execution Evidence & Quality Metrics

OneLogin – Single Sign-On Rollout Across a University's Applications

Comprehensive verification deliverable adhering to IEEE 829 standards. Documents the master verification test suite mapping every functional and non-functional requirement, systematic Boundary Value Analysis (BVA), equivalence partitioning, Role-Based Access Control (RBAC) decision tables, empirical defect tracking logs, Defect Removal Efficiency (DRE), and defect density calculations.

Document Control

Document Title	Software Test Plan & Verification Evidence
Document Version	Version 1.0 (Approved Baseline)
Test Standard Baseline	IEEE Std 829-2008 Standard for Software and System Test Documentation
Traceability Scope	100% Coverage across FR-01 through FR-11 and NFR-01 through NFR-06 (TC-01 to TC-23)
Related Deliverables	SRS Document, UML Design Package, Project Plan, Estimation Sheet, Risk Register & Closure Note

Formal Sign-Off & Approvals

Role	Designation & Stakeholder	Signature	Approval Date
Project Sponsor	Vice-Chancellor	_____	02 Oct 2026
Business Owner	IT Head	_____	02 Oct 2026
QA / Test Lead	Senior Quality Assurance Lead	_____	02 Oct 2026
Technical Lead	Lead Systems Engineer	_____	02 Oct 2026

Quality Exit Gate Policy: Zero Critical or High severity security defects are permitted prior to production deployment. All 23 functional and non-functional requirements must achieve 100% pass status, validating the single sign-on experience and safeguarding student examinations from unexpected system lockouts.

1. Master Verification Suite: Functional Test Cases (TC-01 to TC-13)

Every requirement in the approved IEEE 830 SRS maps directly to an executable test case. Part 1 validates primary authentication, single sign-on across federated applications, single logout, role-based authorization, self-service password reset, and account lockout policies.

ID	Requirement	Test Objective & Method	Expected Result & Acceptance Criteria	Status
TC-01	FR-01 (Auth)	Submit valid credentials to IdP portal	Auth succeeds; sets Secure/HttpOnly session cookie	PASS
TC-02	FR-01 (Auth)	Submit invalid user ID or wrong password	Generic error displayed; zero disclosure of invalid field	PASS
TC-03	FR-02 (SSO)	Access ERP, LMS, and Library in sequence	1 login grants seamless access to all 3 with 0 extra prompts	PASS
TC-04	FR-02 (SLO)	Trigger Single Logout from LMS endpoint	Central session destroyed; ERP and Library demand re-login	PASS
TC-05	FR-03 (Cloud)	Validate SAML assertion token for 4 cloud apps	Valid assertion accepted; tampered/expired signature refused	PASS
TC-06	FR-04 (In-House)	Access in-house LMS directly; verify forms	302 redirects to IdP; direct local login form disabled	PASS
TC-07	FR-05 (Vendor)	Test reverse-proxy header injection on legacy app	Adapter injects credential headers or records exception	PASS
TC-08	FR-06 (RBAC)	Verify role matrix (Student vs. Staff vs. Helpdesk)	Access allowed/denied strictly per role permissions	PASS
TC-09	FR-06 (RBAC)	Admin updates user role in directory	Permission change takes effect across all apps in ≤ 5 min	PASS
TC-10	FR-07 (SSPR)	Complete self-service reset end-to-end	OTP sent; new password updates directory with 0 helpdesk aid	PASS
TC-11	FR-07 (SSPR)	Replay used OTP or submit after 15 min	OTP rejected; error prompts user to request new code	PASS
TC-12	FR-08 (Lockout)	Submit 5 consecutive incorrect passwords	Account locked for 15 minutes; correct pass rejected during lock	PASS
TC-13	FR-08 (Lockout)	Helpdesk agent unlocks locked account via console	Lock flag cleared immediately; user logs in successfully	PASS

Functional Baseline Verified: All 13 core functional test cases passed successfully in the integration staging environment, validating seamless single sign-on across cloud and in-house applications without security regressions.

1. Master Verification Suite: Security & NFR Verification (TC-14 to TC-23)

Part 2 validates system idle timeouts, audit trail hygiene, privileged role multi-factor authentication (MFA), and the 6 measurable non-functional requirements (NFRs) including availability, latency, disaster recovery, usability, and coupling.

ID	Requirement	Test Objective & Method	Expected Result & Acceptance Criteria	Status
TC-14	FR-09 (Timeout)	Leave active SSO session idle for 30 minutes	Session ends; subsequent navigation redirects to IdP login	PASS
TC-15	FR-10 (Audit)	Perform login, lock, unlock, and reset events	Events logged with user ID, IP, timestamp, application	PASS
TC-16	FR-10 (Audit)	Scan audit log storage for sensitive fields	Passwords and OTP reset codes are never written to log sink	PASS
TC-17	FR-11 (MFA)	Helpdesk / Admin login vs. Student login	MFA enforced for privileged roles; bypassed for students	PASS
TC-18	NFR-01 (Uptime)	Synthetic 1-minute HTTP login probe over 30 days	Availability ≥ 99.9% (unplanned downtime ≤ 43.2 min/month)	PASS
TC-19	NFR-02 (Security)	SSL Labs scan & password storage audit	TLS 1.2+ enforced; passwords stored as salted bcrypt hashes	PASS
TC-20	NFR-03 (Latency)	Peak simulated load test (1,000 concurrent users)	Login p95 ≤ 2.0 s; subsequent app launch adds ≤ 1.0 s	PASS
TC-21	NFR-04 (Recovery)	Simulate primary node crash; restore from backup	RTO ≤ 30 minutes; RPO ≤ 15 minutes validated	PASS
TC-22	NFR-05 (Usability)	Moderated usability test with 20 student test users	≥ 90% complete unaided SSPR in ≤ 3 minutes and ≤ 5 steps	PASS
TC-23	NFR-06 (Coupling)	Onboard a test SAML application into staging	0 lines of IdP code changed; pure metadata configuration	PASS

Equivalence Partitioning Summary

- **Credential Validity:** Valid Active Users (Partition 1: Authenticate → 200 OK); Non-existent Users (Partition 2: Generic Auth Failure); Inactive / Suspended Users (Partition 3: Contact Admin Notice).
- **Session States:** Active Interaction (< 30m idle: Token Touched); Expired Session (≥ 30m idle: 302 to IdP); Terminated Session (SLO Triggered: All SP cookies purged).
- **Role Access Partitions:** Student Academic Pool (LMS, Library, Hostel); Staff Operational Pool (ERP, HR, Finance); Admin Security Pool (RBAC Console, Unlock Desk, Audit Sink).

2. Boundary Value Analysis (BVA) & Edge Value Verification

Boundary value analysis targets edge values around critical operational thresholds specified in the SRS: failed-login count (FR-08), session idle timer (FR-09), OTP token lifetime (FR-07), and password complexity.

2.1 Failed-Login Lockout Threshold (FR-08: Target = 5 attempts)

Test ID	Failed Attempt Value	Boundary Classification	Observed System Behavior	Result
BVA-08-01	Attempts = 3	Interior Normal (Valid)	Login denied; warning counter = 3; account remains active	PASS
BVA-08-02	Attempts = 4	Boundary Just Below (Valid)	Login denied; warning: "1 attempt remains before lockout"	PASS
BVA-08-03	Attempts = 5	Exact Boundary (Critical)	Account locked for 15 minutes; audit security event emitted	PASS
BVA-08-04	Attempts = 6	Boundary Just Above (Lock)	Attempt blocked immediately; lockout timer not reset	PASS
BVA-08-05	Successful Login at Count = 4	Reset Equivalence Partition	Auth succeeds; fail counter reset from 4 back to 0	PASS

2.2 Session Idle Timeout (FR-09: Target = 30 minutes)

Test ID	Inactivity Elapsed	Boundary Classification	Observed System Behavior	Result
BVA-09-01	29 minutes 00 seconds	Boundary Just Below (Active)	HTTP request touches session; idle counter reset to 0	PASS
BVA-09-02	29 minutes 59 seconds	Boundary Immediately Below	Session remains valid; target resource renders normally	PASS
BVA-09-03	30 minutes 00 seconds	Exact Expiry Boundary	Session cookie invalidated; IdP session marked EXPIRED	PASS
BVA-09-04	30 minutes 01 seconds	Boundary Just Above (Expired)	HTTP 302 redirect to IdP login with expired session notice	PASS

2.3 SSPR OTP Validity & Password Length (FR-07: 15 min OTP, ≥10 chars)

Test ID	Parameter Tested	Input Value	Observed System Behavior	Result
BVA-07-01	OTP Age Window	14 minutes 59 seconds	Code accepted; user advances to password change screen	PASS
BVA-07-02	OTP Age Window	15 minutes 01 seconds	Code rejected: "One-time code has expired. Request new OTP"	PASS
BVA-07-03	Password Length	9 characters (e.g., "Abcd1234!")	Rejected: "Password must be at least 10 characters"	PASS
BVA-07-04	Password Length	10 characters (e.g., "Abcd12345!")	Accepted; directory hash updated; all active sessions revoked	PASS

3. Decision Tables: Role-Based Access Rules (FR-06)

The decision table systematically models the Role-Based Access Control (RBAC) policy across all 5 user classes, verifying that privilege boundaries are rigidly enforced across academic, administrative, and security operations.

Rule Condition / Action	R1: Student	R2: Staff	R3: Helpdesk	R4: App Owner	R5: IT Admin
Conditions (Inputs / Actions Attempted)					
Is Authenticated via OneLogin IdP?	Yes	Yes	Yes	Yes	Yes
Target: Student Apps (LMS, Hostel, Library)	Allowed	Allowed	Allowed	Allowed	Allowed
Target: Staff Apps (ERP Payroll, Faculty HR)	Denied	Allowed	Denied	App-Specific	Allowed
Action: Self-Service Password Reset	Allowed	Allowed	Allowed	Allowed	Allowed
Action: Unlock Another User's Account	Denied	Denied	Allowed	Denied	Allowed
Action: Trigger Verified Reset for Others	Denied	Denied	Allowed	Denied	Allowed
Action: Assign / Modify User Roles	Denied	Denied	Denied	Denied	Allowed
Action: View Central Audit Logs	Denied	Denied	Lock Events	Own App Only	All Logs
Action: Register / De-register Applications	Denied	Denied	Denied	Request Only	Allowed
Expected Outcomes & Verification Evidence					
HTTP Status on Permitted Action	200 OK	200 OK	200 OK	200 OK	200 OK
HTTP Status on Unauthorized Action	403 Forbidden	403 Forbidden	403 Forbidden	403 Forbidden	403 Forbidden
Multi-Factor Authentication (FR-11)	Bypassed	Bypassed	Enforced	Bypassed	Enforced
Test Execution Status	PASS (TC-08)	PASS (TC-08)	PASS (TC-08)	PASS (TC-08)	PASS (TC-08)

Access Control Security Invariants

- **Least Privilege Enforcement:** Students cannot access ERP payroll endpoints or view peers' records under any circumstance. Attempts trigger an HTTP 403 Forbidden and generate an audit security alarm.
- **Privilege Separation:** IT Helpdesk agents can unlock locked accounts and initiate verified resets, but are strictly prohibited from viewing full audit trails or altering role memberships.
- **Role Propagation SLA:** Verified by TC-09, when an administrator grants staff access to a promoted user, the RBAC cache updates across all federated application tokens within **3.4 minutes** (satisfying the ≤ 5 minute requirement).

4. Empirical Defect Log, Defect Density & DRE

Quality assurance tracked all anomalies discovered across development, integration testing, pilot UAT, and production warranty. The resulting metrics evaluate engineering effectiveness and verification rigor.

4.1 Master Defect Discovery & Resolution Log

Bug ID	Severity	Root Cause & Defect Description	Phase Detected	Phase Resolved	Status
BUG-01	CRITICAL	In-house LMS retained legacy cookie endpoint, allowing local authentication bypass	Integration QA	Integration QA	Closed
BUG-02	HIGH	Single Logout failed to invalidate SAML session on Cloud ERP endpoint	Integration QA	Integration QA	Closed
BUG-03	Medium	Login error returned "Username not found", violating FR-01 generic error rule	Security Scan	Security Scan	Closed
BUG-04	Medium	Session timeout notice displayed raw UTC epoch timestamp rather than local time	Pilot Cohort UAT	Pilot Cohort UAT	Closed
BUG-05	Medium	SSPR SMS gateway throttled during 50-user concurrent verification stress drill	Pilot Cohort UAT	Pre-Cutover	Closed
BUG-06	Low	Mobile viewport clipped application launcher icons on screens under 380px	Production Pilot	Warranty Buffer	Closed

4.2 Quantitative Quality Metrics Calculations

Using the approved Option A engineering baseline effort of **136 person-days** and the defect discovery data, defect density and removal efficiency are calculated as follows:

1. Defect Density Calculation:

Total Defects Detected = 6 defects (BUG-01 through BUG-06)

Total Project Work Effort = 136 Person-Days

Defect Density = Total Defects ÷ Total Effort = 6 ÷ 136 = **0.0441 Defects / Person-Day**

(Benchmark comparison: well within industry quality baseline of ≤ 0.10 defects/p-d)

2. Defect Removal Efficiency (DRE) Calculation:

Internal Defects Detected Pre-Release (E) = 5 defects (BUG-01 to BUG-05)

Post-Release / Production Warranty Defects (D) = 1 defect (BUG-06)

DRE = [E ÷ (E + D)] × 100 = [5 ÷ (5 + 1)] × 100 = **83.33%**

(Quality target achieved: DRE exceeds the mandatory IEEE 829 threshold of ≥ 80%)

Quality Assurance Verdict: The OneLogin SSO implementation satisfies all quality exit gates. With **100% test case pass rate** across TC-01 through TC-23, zero open critical defects, a defect density of **0.044**, and a DRE of **83.33%**, the system is fully approved for university-wide semester deployment.